

Your grade: 100%

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Next item →

1. What is the correct syntax for implementing  $e^{At}$  in Octave if  $A$  is a matrix,  $t$  is a scalar, and  $x_0$  is a vector?

1 / 1 point

- ☐ `exp (A*t)`
- ☐ `expm (A*t) *x0`
- ☐ `exp (A*t) *x0`
- ☒ `expm (A*t)`

✔ **Correct**  
Yes, this is correct.

2. The forced-output response  $z(t)$  of a state-space model has which component(s)? (Select all that apply.)

1 / 1 point

- ☒ A response component due to initial conditions  $x(0)$ .

✔ **Correct**  
Yes, this is correct.

- ☒ A response component due to the forcing input  $u(t)$  via a convolution computation.

✔ **Correct**  
Yes, this is correct.

- ☒ A response component due to the forcing input  $u(t)$  via a direct feedthrough term.

✔ **Correct**  
Yes, this is correct. Note that not all models will have a nonzero  $D$  matrix, but the most general form of state-space model does allow for the direct-feedthrough term.

- ☐ A response component due to the frequencies present in  $u(t)$  via a transfer-function operation.

3. How can you compute  $e^{At}$  by hand (i.e., without relying on Octave or MATLAB or some similar engineering software tool.) if  $A$  is a matrix and  $t$  is a scalar? (Select all that apply.)

1 / 1 point

- ☐ You cannot compute  $e^{At}$  by hand.
- ☒ By evaluating  $e^{At} = \mathcal{L}^{-1} \left( (sI - A)^{-1} \right)$ .

✔ **Correct**  
Yes. This is correct.

- ☐ By evaluating  $e^{At} = I + At + A^2t^2/2! + \dots$ .
- ☐ By computing  $\exp(a_{i,j}t)$  for every element  $a_{i,j}$  of the matrix  $A$ .